PHYSICS WALLAH

POLYMERS

ADDITION POLYMERS

POLYTHENE
[ETHYLENE]
$CH_2=CH_2$

- **Low density polythene**
  - High pressure of 1000 to 2000 atm at a temperature of 350K to 570 K
  - Presence of traces of dioxygen or a peroxide initiator
  - Highly branched structure
  - Chemically inert & tough but flexible Poor conductor of electricity
  - Used in the insulation of electricity carrying wires and manufacture of squeeze bottles, toys & flexible pipes
- **High density polythene**
  - Low pressure of 6-7 atm and a temperature of 333 K to 343 K
  - Presence of a catalyst such as triethylaluminium and titanium tetrachloride (Ziegler-Natta catalyst
  - Linear polymers
  - Chemically inert and more tough and hard
  - Used in the manufacturing buckets, dustbins, bottles, pipes, etc

POLYPROPYLENE
[PROPYLENE]
$CH_2=CHCH_3$
Ropes, Toys, Pipes, Fibres, etc.

POLY VINYL CHLORIDE (PVC)
[VINYL CHLORIDE]
$CH_2=CHCl$
Rain coats, Hand bags, Vinyl flooring

POLY STYRENE
[STYRENE]
$CH_2=CHC_6H_5$
Insulator, wrapping material,

POLY ACRYLONITRILE
[PAN, ORLON, ACRILAN]
[ACRYLONITRILE]
$CH_2=CH-CN$
blankets, clothing

POLY TETRAFLUORO ETHYLENE (TEFLON)
[TETRAFLUORO ETHYLENE]
$CF_2=CF_2$
Non-stick surface coated utensils

POLY METHYL METHACRYLATE
[PMMA, Lucite, Plexiglas]
[METHYL METHACRYLATE]
$CH_2=C(CH_3)\ CO_2CH_3$

CLASSIFICATION OF POLYMERS

Based on molecular forces
- Elastomers: Rubber like solids with elastic property eg: Buna-S, Buna-N, Neoprene
- Fibres: High tensile strength & high modules eg: Polyamides (nylon-6,6) Polyesters (terylene)
- Thermoplastic polymers: Intermediate b/w elastomer & fibres Capable of repeat softening on heating & hardening on cooling eg: Polythene, polystyrene, polyvinyl etc.
- Thermosetting polymers: Crosslinked & heavily branched polymers. These cannot be reused eg: Bakelite, resins, urea-formaldehyde etc.

Based on structure
- Linear eg: high density polythene, poly vinyl chloride
- Branched eg: low density polythene
- Cross-linked eg: Bakelite, melamine - formaldehyde resin

Based on source
- Natural polymers eg: protein, cellulose, starch, resins, rubber
- Semi-Synthetic polymers eg: Cellulose acetate (rayon), cellulose nitrate etc.
- Synthetic polymers eg: Nylon-6, 6, Buna-S

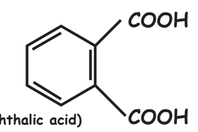
Based on mode of polymerisation
- Addition polymers
- Condensation polymers Eg: Nylon 6,6

Types of Polymerisation Reactions
- Addition Polymerisation or Chain Growth Polymerisation
- Condensation Polymerisation or Step Growth Polymerisation

CONDENSATION POLYMERISATION

POLYESTERS

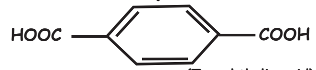
Glyptal



(phthalic acid)
+
$HO - CH_2 - CH_2 - OH$
(Glycol)

Paints & Lacquers

Dacron or Terylene



(Terephthalic acid)
+
$HO - CH_2 - CH_2 - OH$
(Glycol)

Crease resistant, used in blending with cotton & wool fibres, glass reinforcing materials in safety helmets, etc

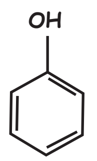
Poly β-hydroxybutyrate-co-β-hydroxy valerate (PHBV)

$CH_3-CH(OH)-CH_2-COOH + CH_3-CH_2-CH(OH)-CH_2-COOH$
(3 - hydroxy butanoic acid) (3 - hydroxy pentanoic acid)

PHBV is used in speciality packaging, orthopaedic devices and in controlled release of drugs

RESINS

Phenol - Formaldehyde polymer

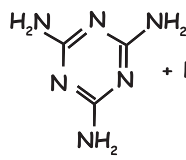


$+ CH_2O$

Novolac used in paints

- Novolac on heating with formaldehyde undergoes cross linking to form an infusible solid mass called bakelite
- Electrical switches and handles of various utensils

Melamine - Formaldehyde polymer



$+ HCHO$

Manufacture of unbreakable crockery

Urea-Formaldehyde polymer
$NH_2-CO-NH_2 + HCHO$

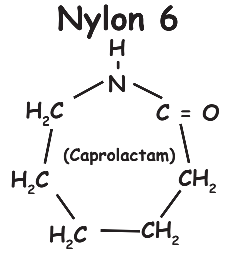
For making unbreakable cups & laminated sheets

POLYAMIDES

Nylon 6,6
$nHOOC(CH_2)_4COOH + n\ H_2N\ (CH_2)_6\ NH_2$
(Adipic acid) (Hexa methylene diamine)

Making sheets, bristles for brushes & in textile industry.

Nylon 6



(Heating with H_2O)

Manufacture of tyre cords, fabrics and ropes

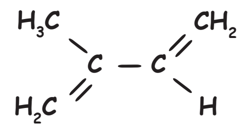
Nylon 2 - Nylon 6
$H_2N - CH_2 - COOH$
(Glycine)
+
$H_2N - (CH_2)_5 - COOH$
(Amino caproic acid)

RUBBER

1. NATURAL RUBBER

1. Natural rubber

(Isoprene)
(2-methyl-1, 3-butadiene)



Vulcanisation of rubber:

Heating a mixture of raw rubber with sulphur and an appropriate additive at a temperature range between 373K to 415 K.

2. SYNTHETIC RUBBER

Neoprene

(Chloroprene)
(2-Chloro-1, 3-butadiene)

$CH_2 = C(Cl) - CH = CH_2$

Manufacturing of conveyor belts, gaskets & hoses

Buna - N

1,3-Butadiene+Acrylonitrile

$CH_2 = CH - CH = CH_2 + CH_2 = CH(CN)$

Making of oil seals, tank lining, etc

Buna - S

1, 3 - Butadiene + Styrene

$CH_2=CH-CH=CH_2+C_6H_5CH=CH_2$

Auto tyres, floor tiles, foot wear, cable insulation